

ESTIMATING THE EFFECT OF AI-TOOL USE ON HOURS SPENT ON SCHOOLWORK: A HYPOTHESIS TEST AND A BACK-DOOR ADJUSTMENT

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ABSTRACT. Generative AI tools have been adopted rapidly in education since the public release of ChatGPT in late 2022. We ask what effect AI-tool use X has on the number of hours students spend on independent schoolwork Y , and we show that the answer depends more on what is adjusted for than on the choice of test statistic. Two samples are analysed. The primary sample is a convenience survey of 15 Ngee Ann Polytechnic students collected in July 2026. Splitting it at the sample median of self-reported AI hours and applying a pooled two-sample t -test gives $t = -1.4944$ on 13 degrees of freedom, two-tailed $p = 0.1590$, so equality of means is not rejected at $\alpha = 0.05$. The decision survives Welch’s test ($p = 0.1909$), an exact permutation test ($p = 0.1535$), the Mann–Whitney test ($p = 0.3357$) and deletion of the most influential observation ($p = 0.3575$). It is nevertheless uninformative rather than null: the 95% confidence interval for the mean difference is $(-17.47, 3.18)$ hours per week, the smallest difference the design could detect with probability 0.80 is roughly 14.5 hours per week, and removing one respondent halves the point estimate. The test also conditions on the empty set, which does not satisfy Pearl’s back-door criterion for these variables, so it measures an unadjusted association. To show that this matters, we repeat the question on a public secondary dataset carrying the confounders our own instrument omitted, the Colombian study-habits survey ($n = 357$). Under an adjustment set that satisfies the criterion the estimate is $+0.216$ log-odds ($SE = 0.132$, $p = 0.102$), equivalently $+0.159$ hours per week ($SE = 0.094$, $p = 0.092$); the unadjusted estimate on the same data is $+0.088$ hours ($p = 0.360$). Valid adjustment nearly doubles the estimate, and adjusting instead for a post-treatment mediator collapses it by 53%. Both samples point away from the substitution hypothesis and neither reaches significance. Throughout we report an adjusted association, not a demonstrated causal effect.

1. INTRODUCTION

Generative AI tools have spread through education quickly since ChatGPT was released publicly in late 2022, and the obvious question for a student is whether using them changes how long schoolwork takes. Two mechanisms are proposed in the literature and they point in opposite directions.

Under *substitution*, AI absorbs work the student would otherwise have done alone, so heavier use goes with fewer independent study hours. Under *complementarity*, AI raises the productivity of study time and is adopted preferentially by students already investing heavily, so heavier use goes with more hours. Existing work supports the first mechanism at the level of a single task without settling the aggregate. Dahri *et al.* [5] report that ChatGPT use predicts perceived reductions in time spent on traditional learning resources, but their outcome is a perception item about resource substitution rather than total schoolwork hours. Li *et al.* [11] log GenAI-assisted problem solving in fine detail for 279 students, and their timestamps measure time on a two-week assignment rather than a student’s weekly total. Contrac-tor and Reyes [4] survey 634 students and collect weekly study hours alongside GenAI frequency, which is the closest variable match available, but respondent-level microdata is not distributed with the paper. Task-level time savings are therefore better evidenced than any net effect on hours, and the net effect is what a student actually experiences.

The question studied here is: *what is the effect of AI-tool use X on hours spent on independent schoolwork Y ?* Written as an interventional quantity it is $P(y \mid do(x))$, not $P(y \mid x)$. The two differ whenever a common cause of X and Y is left unadjusted, and at least three such causes are plausible here: workload and competing time demands, prior ability and motivation, and course difficulty. Pearl’s back-door criterion [16, 17] states when the gap can be closed from observational data, and Cinelli *et al.* [3] catalogue the ways in which adding the wrong variable widens it instead.

Concretely, this study sets out to (i) measure weekly AI-tool use and independent study time among Ngee Ann Polytechnic students through a primary survey ($n = 15$); (ii) test whether mean independent study time differs between low and high AI-use students, using a two-sample t -test at $\alpha = 0.05$, together with the assumption checks, interval estimate and error-rate analysis that the decision requires; and (iii)

establish the limits of that conclusion — sampling bias, measurement error and unadjusted confounding under the back-door criterion — so that the finding is reported as an association rather than a causal effect.

Section 2 reviews the empirical literature on AI use and academic time. Section 3 states what the estimand requires of the data, describes the survey instrument, documents the processing of all nine items including two defects that changed the numbers, and sets out the sampling model. Section 4 formulates the hypotheses, derives and applies the test, reports the interval estimate and the two error rates, examines every assumption the test requires, and then shows why the test cannot answer the causal question no matter how robust the decision is. Section 5 answers that question on a secondary dataset that has the confounders our own instrument lacks, and uses the same dataset to demonstrate the mediator-adjustment error empirically. Section 6 states what the evidence supports.

The arithmetic of a two-sample t -test is elementary. The contribution here is in saying precisely why a non-significant result from fifteen self-selected respondents is inconclusive rather than null, and in showing on real data how much a valid adjustment set moves an estimate that the t -test leaves unadjusted.¹

2. LITERATURE REVIEW

This section synthesises the emerging empirical literature on the pathways through which AI-tool use affects the total hours dedicated to schoolwork.

2.1. Time compression and efficiency gains. A recurring finding is that ChatGPT use reduces the time students need to complete *specific* academic tasks, primarily by compressing information retrieval, writing and problem-solving. Students report accomplishing assignments more rapidly than with traditional methods such as search engines, library research or unaided writing [13, 5]. It is worth marking at the outset that this is a claim about task-level time, not about weekly totals; the two come apart precisely when saved time is reallocated rather than banked.

2.2. Cognitive load reduction. Cognitive Load Theory offers the mechanism. By taking over information management, an AI assistant plausibly reduces extraneous load and frees working memory for germane processing. In a 16-week randomised controlled trial of a ChatGPT-enhanced mobile instant messaging system, 63 graduate students were randomised to the AI-supported condition ($n = 33$) or to conventional messaging ($n = 30$), and the treatment group showed improvements across the behavioural, cognitive and emotional dimensions of engagement [8]. This is the only randomised design in the literature reviewed here, which is why it carries disproportionate weight in what follows.

2.3. Time reallocation toward deeper engagement. Time saved on routine work may be reallocated to more demanding activity rather than producing a net reduction in total schoolwork time. In a quasi-experimental study of 366 undergraduates, Li, Sadiq, Qambar and Zheng [10] found that ChatGPT-integrated instruction was associated with autonomous motivation ($\beta = 0.551$, $p < 0.01$) and engagement ($\beta = 0.506$, $p < 0.01$), which in turn were associated with self-directed learning ($\beta = 0.481$, $p < 0.01$) and improved research skills. The direct path from ChatGPT use to research skills was not significant; the association ran entirely through the mediating variables. On this reading AI does not simply compress time, it changes what the time is spent on — which is exactly the pattern that would produce a null or positive effect on total hours despite genuine task-level savings.

2.4. The risk of reduced independent work time. The countervailing concern is that AI use displaces independent critical thinking and exploratory research. In a qualitative study of learners' experiences, participants reported “surface-level understanding of topics due to overreliance,” “hindrance to the development of independent research skills” and “loss of connection with in-depth learning”; one participant worried that “solely relying on ChatGPT for information retrieval could compromise the depth of understanding and critical analysis necessary in academic research” [13]. A cross-sectional survey of 476 Chinese college students across 67 institutions and 17 provinces similarly recorded concerns about dependency on the tool and about the ethical risks of using it [2].

¹Code and data are available at https://github.com/lohjo/CAEM3A_Screen_Time_Project.

2.5. Limitations of the existing literature. Four limitations bear directly on the present study. First, most studies rely on self-reported time estimates rather than objective tracking. Second, cross-sectional and quasi-experimental designs preclude strong causal inference; only one study reviewed here is randomised [8]. Third, the settings are predominantly higher education, limiting generalisability. Fourth, and most important for us, few studies measure *total hours spent on schoolwork* as a primary outcome — most measure efficiency, perceived usefulness or learning outcomes and infer time implications secondarily.

Taken together, the evidence establishes that AI tools reduce the time required for specific academic tasks, and leaves the net effect on total schoolwork hours genuinely open. That net effect appears contingent on the quality of engagement, the presence of instructor guidance, the nature of the task and the student’s self-regulatory capacity. Under favourable conditions saved time is reallocated to higher-order work; under unfavourable conditions AI enables shortcut behaviour that bypasses processing altogether. Note that the causal language in this paragraph is warranted only for the task-level claim, which has randomised support; the aggregate claim is the one this paper tests, and it is the one the literature does not settle.

3. DATA AND ASSUMPTIONS

3.1. What the estimand requires. For a back-door adjustment set Z to exist and be usable, the instrument must supply four things: a quantitative or ordinal measure of AI use X ; a quantitative measure of total study time Y , not time on a single task and not a Likert perception item; at least one pre-treatment measure of competing time demands W ; and at least one pre-treatment measure of ability, motivation or course difficulty A . Variables caused by X , such as time saved or perceived academic benefit, must be measurable but must *not* enter Z when the total effect is wanted.

Our own survey supplies X and Y and neither W nor A . That gap is the reason Section 5 exists, and it is a design failure we record rather than repair.

3.2. The instrument. The instrument was a Google Form of nine items, circulated between 11 and 21 July 2026 within the researchers’ own course and social networks. Fifteen students responded. Table 1 lists the items and their analytic role.

#	Item	Variable	Role
1	Current year of study	year	Background
2	School / course cluster	school	Background
3	Hours per week using AI tools (e.g. ChatGPT, Claude, Gemini) for schoolwork	ai_hours	Explanatory X
4	Hours per week of independent study time (e.g. no AI tools)	study_hours	Response Y
5	AI tool(s) used most (select all)	tools	Construct check
6	Primary use of AI tools (select one)	use	Construct check
7	Academic confidence, 1–10	confidence	Intended covariate
8	“I would struggle academically without AI tools”, 1–5	dependence	Intended covariate
9	Submission timestamp (automatic)	ts	Administrative

TABLE 1. Survey instrument. Items 3 and 4 requested a plain number and supplied a worked example (“e.g. 3.5”) to discourage range answers.

3.3. Sampling method and its consequences. This is a convenience sample and not a probability sample. The distinction is not a formality. The sampling frame is unknown, no respondent has a calculable inclusion probability, and the guarantee that a sample mean is unbiased for a population mean comes from the sampling design, which is absent. The realised composition shows the recruitment route: six of fifteen respondents are from one school (SOE) and six are in Year 3, against roughly five per year for an evenly spread sample of a three-year diploma. Self-selection compounds this, since an optional survey about AI most attracts students with strong views about AI in either direction. The sign of the resulting non-response bias cannot be determined, because enthusiasts over-reporting use and sceptics over-reporting study time push the association in opposite directions. Nothing in the analysis repairs any of this, so no statement below is generalised beyond the fifteen respondents observed.

3.4. Processing the responses. Four decisions were required before analysis, and two of them changed reported numbers.

- *Blank cells.* Four of the fifteen respondents left exactly one field blank each: one skipped year, one school, one the primary-use item, and one the dependence rating. Both `ai_hours` and `study_hours` are complete for all fifteen, so the test in Section 4 uses the full sample with no deletions and the blanks restrict only subgroup work.
- *An invalid category.* One respondent selected “Year 0”, which is not a year of study at this institution. The row is retained because both hour responses are usable, and the respondent’s year is treated as unknown rather than imputed.
- *A parsing defect.* One respondent answered “None” to the tools item. Read with default settings, `pandas` coerces the string `None` to a missing value, which converts a substantive answer into a hole and inflates the blank count from four to five. Re-reading with `keep_default_na=False` restores it. “None” is a legitimate answer to a tool-use question and the default configuration destroys it; the earlier draft of this analysis reported one missing value in `tools` for exactly this reason, and the correct count is zero.
- *Free text in closed fields.* Three respondents typed prose into the multiple-choice tools item (“Copilot cuz Its built in”, “None, Deepseek sometimes”, “None, If really needed then Google AI overview”) and one answered the primary-use item with a description of group work under time pressure. These were not recoded into existing categories, because recoding would conceal the finding, which is that the option list did not span the respondents’ actual behaviour. They are counted separately in Table 2. The raw export was not modified; all cleaning operates on a copy.

3.5. All nine items. Table 2 summarises the categorical items and Table 3 the quantitative and ordinal ones.

Item	Category	<i>n</i>
Year of study	Year 1 / Year 2 / Year 3	2 / 5 / 6
	“Year 0” (invalid) / blank	1 / 1
School cluster	SOE / HS / HMIS	6 / 3 / 2
	BA / ICT / LSCT / blank	1 / 1 / 1 / 1
Tools (24 mentions, 15 respondents)	ChatGPT / Gemini / Claude	8 / 8 / 4
	DeepSeek / Copilot / Google AI Overview	2 / 1 / 1
	answered “none” then named a tool	3
Primary use	Solving problems / Explaining concepts	5 / 4
	Checking or writing work / Translation or summarising	2 / 2
	Free text / blank	1 / 1

TABLE 2. Categorical items. Three respondents answered “none” to the tools item and then named a tool, indicating that “AI tool use” was not understood uniformly; those respondents are likely to under-report `ai_hours`.

Variable	<i>n</i>	Mean	Median	SD	Min	Max
<code>ai_hours</code> (h/week)	15	3.02	2.00	2.61	0	8
<code>study_hours</code> (h/week)	15	10.33	6.00	9.63	2	37
<code>confidence</code> (1–10)	15	5.40	6.00	2.35	1	8
<code>dependence</code> (1–5)	14	2.29	2.50	1.14	1	4

TABLE 3. Quantitative and ordinal items. Two respondents reported exactly 0 AI hours and a third 0.25, so the sample contains genuine non-users. The standard deviation of `study_hours` is almost as large as its mean, which is the main reason the test in Section 4 is inconclusive.

Three pairwise relationships are worth recording, none significant at 5% and none of which could plausibly have been at $n = 15$. The relationship this paper is about, taken on the original continuous scale of X , is positive: Pearson $r = 0.358$ ($p = 0.191$; Spearman $\rho = 0.360$, $p = 0.188$). More AI use

goes with *more* independent study, not less, which is the direction the median-split test in Section 4 also finds while discarding information the correlation retains.

Respondents reporting more AI hours rated their academic confidence lower, $r = -0.432$ ($p = 0.108$). Cross-sectional data cannot separate the two readings of this: low confidence plausibly drives students to lean on AI, and heavy reliance on AI plausibly erodes confidence. Both fit equally well, and the ambiguity is the subject of Section 4.7.

The **dependence** item, which was collected in order to serve as a covariate, tracks confidence weakly (Spearman $\rho = 0.239$, $p = 0.411$) and AI hours not at all ($\rho = -0.033$, $p = 0.911$). Whatever it measures, it is not a usable proxy for prior ability, and Section 4.7 gives a second and stronger reason not to adjust for it.

3.6. Measurement. Both X and Y are self-reported retrospective estimates of weekly hours, which carries three problems. Recall error is expected to be large, because respondents reconstruct diffuse activity such as “studying” poorly. Social-desirability pressure plausibly acts in opposite directions on the two items, understating AI use, which may feel like an admission of shortcutting, and overstating study time; the net effect on the observed association cannot be signed. Definitional ambiguity is the sharpest of the three: item 4 asks for study time using “no AI tools,” but a student who consults a chatbot intermittently during a three-hour session has no principled way to classify those hours, and the three respondents in Table 2 who answered “none” and then named a tool demonstrate that the construct was not read uniformly.

The largest observation, 37 hours per week of independent study, is a plausible unit error (a day read as a week) or a misreading of the item rather than a genuine value. With the survey closed it cannot be verified, so it is retained in the primary analysis and its influence is quantified in Section 4.6 instead of being assumed away.

3.7. The sampling model.

Assumption 1. Conditionally on the observed vector of AI hours, and hence on the induced partition into groups of sizes n_1 and n_2 , the response values satisfy

$$Y_{1,1}, \dots, Y_{1,n_1} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_1, \sigma^2), \quad Y_{2,1}, \dots, Y_{2,n_2} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_2, \sigma^2),$$

with the two samples mutually independent, $\sigma^2 \in (0, \infty)$ unknown and common to both groups, and $\mu_1, \mu_2 \in \mathbb{R}$ unknown.

Assumption 1 bundles four requirements: independence within each group, independence between groups, normality of each group, and homoscedasticity. Independence between groups holds by construction, since each respondent contributes to exactly one group. Independence within groups is a design claim that convenience sampling makes debatable, since respondents recruited through overlapping social and course networks may share courses, deadlines and study habits, and it is the one assumption for which no diagnostic is available. The remaining two are testable and both are examined in Section 4.6, where neither survives comfortably.

4. HYPOTHESIS TEST AND RESULTS

4.1. Reduction to a two-sample comparison. The methods available here test a single population mean or a difference of two means, so the bivariate question is brought into that framework by dichotomising X and comparing the resulting groups on Y . The sample median of **ai_hours** is 2.0 hours per week and four respondents report exactly that value, so the treatment of ties fixes the group sizes: assigning them to the lower group gives $n_1 = 8$ and $n_2 = 7$, assigning them to the upper group gives 4 and 11. The balanced allocation was chosen, since 8 versus 7 has more power, and the rule is a function of X alone and so cannot be steered by the outcome. Mean AI use is 1.16 h/week in the low group and 5.14 h/week in the high group. Remark 2 states what this choice costs.

4.2. Hypotheses. Write μ_1 for the mean weekly independent study time in the population of low AI-use students and μ_2 for the corresponding mean among high AI-use students. The hypotheses are

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 \neq \mu_2,$$

tested two-tailed at $\alpha = 0.05$. No directional prediction was registered before the data were inspected, and substitution and complementarity predict opposite signs, so selecting a one-tailed alternative after seeing which way the sample means fell would inflate the true type I error rate above its nominal level.

A pooled two-sample t -test [19] is used. The population variances are unknown and must be estimated, which rules out a z -test; both groups are small ($n_1 = 8$, $n_2 = 7$), so the central limit theorem does not

deliver approximate normality of the standardised difference; and the two groups are independent by construction.

4.3. The null distribution. Define the group means $\bar{Y}_1, \bar{Y}_2$, the group variances S_1^2, S_2^2 , the pooled variance and the statistic

$$(1) \quad S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}, \quad T = \frac{\bar{Y}_1 - \bar{Y}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

Theorem 1. *Under Assumption 1, with $\nu = n_1 + n_2 - 2$,*

$$T \sim t_\nu(\delta), \quad \delta = \frac{\mu_1 - \mu_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

In particular, under $H_0 : \mu_1 = \mu_2$ one has $\delta = 0$ and $T \sim t_\nu$ exactly, for every value of the unknown σ^2 .

Proof. By the standard sampling theory of the normal distribution, within each group $\bar{Y}_i$ and S_i^2 are independent, with $\bar{Y}_i \sim \mathcal{N}(\mu_i, \sigma^2/n_i)$ and $(n_i - 1)S_i^2/\sigma^2 \sim \chi_{n_i-1}^2$. The two samples are independent, so $\bar{Y}_1, \bar{Y}_2, S_1^2, S_2^2$ are mutually independent. Hence $Z := (\bar{Y}_1 - \bar{Y}_2)/(\sigma\sqrt{1/n_1 + 1/n_2}) \sim \mathcal{N}(\delta, 1)$, and $V := \nu S_p^2/\sigma^2$ is a sum of two independent chi-square variables on $n_1 - 1$ and $n_2 - 1$ degrees of freedom, so $V \sim \chi_\nu^2$ with $Z \perp V$. Dividing numerator and denominator of (1) by σ gives $T = Z/\sqrt{V/\nu}$, which is the defining representation of $t_\nu(\delta)$. Since σ cancels, the law of T does not depend on it. $\square$

Theorem 1 is what licenses the critical value: the rejection region $\{|T| > t_{\nu, 1-\alpha/2}\}$ has probability exactly α under H_0 , not approximately α , provided Assumption 1 holds.

It also settles a worry raised by the design. Because the grouping rule is a measurable function of the vector of AI hours alone, conditioning on that vector fixes the partition and the group sizes as constants; under the sharp null that Y is independent of X the responses remain i.i.d. normal with common mean and variance regardless of the labels, so Assumption 1 holds conditionally with $\mu_1 = \mu_2$, and averaging the exact conditional level over X preserves it. The median split does not disturb the type I error rate.

Remark 2. *It does disturb the interpretation. Because the cut point is the sample median, μ_1 and μ_2 are means of subpopulations whose definition varies from sample to sample, so $\mu_1 - \mu_2$ is not a fixed feature of any population. Dichotomising also discards the ordering within each group, treating a respondent at 2 hours and one at 0 hours as identical, which is why the median split has less power than the correlation reported in Section 3 on the same data. The tie allocation is defensible on the stated ground that 8 versus 7 beats 4 versus 11, and it depended only on X , but it was fixed after the data were collected rather than in advance.*

4.4. Computation and decision. The observed data are given in Table 4.

Group	n	Observed study_hours (h/week)	Mean	SD	Variance
Low AI use (≤ 2 h)	8	2, 3, 5, 5, 6, 7, 8, 20	7.0000	5.6061	31.4286
High AI use (> 2 h)	7	3, 5, 5, 10, 19, 20, 37	14.1429	12.1714	148.1429

TABLE 4. Group summaries.

The observed difference in means is $\bar{y}_1 - \bar{y}_2 = 7.0000 - 14.1429 = -7.1429$ hours per week. The pooled variance is

$$s_p^2 = \frac{7(31.4286) + 6(148.1429)}{13} = \frac{220.0000 + 888.8571}{13} = 85.2967, \quad s_p = 9.2356,$$

the standard error is

$$SE = s_p \sqrt{\frac{1}{8} + \frac{1}{7}} = 9.2356 \sqrt{0.267857} = 9.2356 \times 0.517549 = 4.7799,$$

and therefore

$$t = \frac{-7.1429}{4.7799} = -1.4944, \quad \nu = 8 + 7 - 2 = 13.$$

The two-tailed critical value is $t_{13, 0.975} = 2.1604$, so H_0 is rejected if $t < -2.1604$ or $t > 2.1604$. The observed value lies inside the acceptance region, and the corresponding p -value is

$$p = 2\mathbb{P}(T_{13} > 1.4944) = 0.1590 > 0.05.$$

Decision: do not reject H_0 . At the 5% level there is insufficient evidence to conclude that mean weekly independent study time differs between low and high AI-use students.

4.5. Interval estimate and the two error rates.

Corollary 3. *Under Assumption 1, $(\bar{Y}_1 - \bar{Y}_2) \pm t_{\nu, 1-\alpha/2} S_p \sqrt{1/n_1 + 1/n_2}$ is a $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$. Here it equals $-7.1429 \pm 2.1604 \times 4.7799$, that is $(-17.47, 3.18)$ hours per week.*

Proof. By Theorem 1, shifting the numerator by the true mean difference makes the noncentrality zero, so $((\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)) / (S_p \sqrt{1/n_1 + 1/n_2}) \sim t_\nu$ for all parameter values. This pivot lies in $[-t_{\nu, 1-\alpha/2}, t_{\nu, 1-\alpha/2}]$ with probability $1 - \alpha$, and rearranging for $\mu_1 - \mu_2$ gives the interval. $\square$

The interval is the decisive quantity. It is compatible with low AI users studying up to 17.5 hours per week more than high users, and simultaneously with high users studying up to 3.2 hours per week more. It excludes essentially nothing of practical interest, so the failure to reject reflects the width of the interval and not the smallness of the effect.

Type I error. By Theorem 1 the probability of rejecting a true H_0 is exactly $\alpha = 0.05$ under Assumption 1, and the median split does not disturb it. This is the only error rate the design controls.

Type II error, expressed as a minimum detectable effect. Nothing controls the type II error rate β , and the cleanest way to see how badly is to ask what the design was capable of detecting. Holding $n_1 = 8$, $n_2 = 7$, $\alpha = 0.05$ and using $s_p = 9.2356$ as an estimate of σ , the smallest true difference detectable with probability 0.80 solves $\mathbb{P}(|t_{13}(\delta)| > 2.1604) = 0.80$, which gives $\delta \approx 3.03$ and hence

$$|\mu_1 - \mu_2| = \delta \sigma \sqrt{\frac{1}{8} + \frac{1}{7}} \approx 3.03 \times 4.7799 \approx 14.5 \text{ hours per week.}$$

That threshold exceeds the low-use group’s entire weekly mean of 7.0 hours and exceeds the whole-sample mean of 10.3 hours. The study could only reliably detect a difference larger than the typical quantity of studying it set out to measure. A difference of, say, 5 hours per week — substantial in context — falls far below the threshold and would therefore be missed far more often than not. Because this quantity depends only on n_1 , n_2 , α and the variability, and not on the result obtained, it is a statement about the design rather than a restatement of the p -value.

Why retrospective power is not offered as evidence. The observed standardised effect is

$$d = \frac{|\bar{y}_1 - \bar{y}_2|}{s_p} = \frac{7.1429}{9.2356} = 0.7734,$$

which is large by conventional standards, and the power of the two-tailed test against a true effect of exactly that size is $\mathbb{P}(|t_{13}(\delta)| > 2.1604)$ with $\delta = d \sqrt{n_1 n_2 / (n_1 + n_2)} = 1.4943$, evaluating to 0.283. This figure is reported for completeness only. Power computed from the observed effect is a strictly decreasing function of the observed p -value and so conveys nothing the p -value has not already conveyed; every non-significant result mechanically yields an observed power below 0.5 [7]. The argument that these data are uninformative rests on Corollary 3 and on the minimum detectable effect above, both of which are computable before any outcome is seen.

Sample size for a future study. Solving the same expression for equal group sizes at $\alpha = 0.05$ and power 0.80 gives $n = 28$ per group against $d = 0.7734$. That target inherits the instability of d , which was itself estimated from fifteen observations, so it should be treated as a floor rather than a design specification. Fixing instead a substantively interesting difference of 5 hours per week in advance, and retaining $s_p = 9.2356$ as the variability estimate, requires roughly 55 per group.

Two consequences follow. “Do not reject” must not be read as “accept”: these data are close to uninformative about H_0 , and absence of evidence is not evidence of absence. And the point estimate should not be reported as a finding, for the reason given at the end of Section 4.6.

4.6. Assumptions examined. Theorem 1 is exact, so the credibility of the p -value rests entirely on Assumption 1. Two of its four components are testable and neither is comfortable.

Normality. The Shapiro–Wilk statistic [18] for the low-AI group is $W = 0.7540$ with $p = 0.0090$, rejecting normality at the 5% level; the group is strongly right-skewed by the single respondent at 20 hours against a cluster between 2 and 8. The high-AI group is not rejected ($W = 0.8636$, $p = 0.1632$), but with $n = 7$ that test has very little power, so its non-rejection is weak evidence of anything.

Homoscedasticity. The high-AI group’s sample variance is 4.71 times the low group’s (148.14 versus 31.43), and the verdict depends on which test is used, as Table 5 shows.

Reporting only the median-centred value, which is the software default, would overstate the case for pooling. All four tests are nearly powerless at these sample sizes, so non-rejection provides no real reassurance in either direction, and the honest reading is that homoscedasticity is unverified rather than supported. Welch’s test [20] is accordingly reported alongside the pooled one.

Test	Statistic	p	Verdict at 5%
Levene, median-centred (Brown–Forsythe) [1]	2.6946	0.1246	does not reject
Levene, mean-centred [9]	4.9098	0.0452	rejects
Bartlett	3.4121	0.0647	does not reject
Variance-ratio F	4.7136	0.0617	does not reject

TABLE 5. Tests for equality of variances, $n_1 = 8$, $n_2 = 7$. Three of the four lie between 0.045 and 0.065.

Independence within groups. No diagnostic is available. Respondents recruited through shared social and course networks may have correlated workloads and study habits, which would deflate the standard error and make the reported p -value anti-conservative. This is a limitation of the recruitment design and cannot be repaired at the analysis stage.

Because the conditions for the exact result are not met, the conclusion is worth reporting only if it survives their relaxation. Four checks are given in Table 6.

Analysis	Assumptions relaxed	Statistic	p
Pooled t (primary)	—	$t = -1.4944$, $\nu = 13$	0.1590
Welch t	equal variances	$t = -1.4260$, $\nu = 8.19$	0.1909
Exact permutation test	normality and equal variances	$ t \geq 1.4944$	0.1535
Mann–Whitney U	normality, equal variances, scale	$U = 19$	0.3357
Delete the 37 h observation	—	$t = -0.9568$, $\nu = 12$	0.3575

TABLE 6. Robustness of the decision. Every analysis fails to reject at $\alpha = 0.05$.

The permutation test deserves comment because it is the only one exact under the stated conditions. Under the sharp null that the group label is uninformative about Y , and conditionally on the pooled multiset of response values and on the group sizes, each of the $\binom{15}{8} = 6435$ assignments of eight values to the low group is equally likely. Enumerating all of them and recomputing T each time gives the exact null distribution of $|T|$ without any distributional assumption; 988 of the 6435 assignments produce $|T| \geq 1.4944$, so $p = 988/6435 = 0.1535$. This is close to the parametric 0.1590, which indicates that the normality failure is not what is driving the result. The Mann–Whitney figure [12] is obtained the same way, by enumerating the 6435 assignments, because the observed values contain ties across groups (the values 3, 5 and 20 each appear in both) and the tabulated exact null distribution assumes none.

Two caveats keep the table from being decisive. The permutation test addresses the sharp null of no distributional difference, so under unequal variances it can reject for reasons unrelated to the means, and the Mann–Whitney test compares stochastic ordering rather than means, so under heteroscedasticity it is not testing H_0 as stated. And these five analyses are not independent confirmations, since they reuse the same fifteen observations; their agreement shows that the decision is insensitive to the choice of assumptions, not that the evidence against H_0 has been checked five times.

The deletion row bears on the outlier discussed in Section 3. Removing the respondent reporting 37 hours per week moves the estimated difference from -7.1429 to -3.3333 and more than doubles the p -value. The decision is stable; the magnitude is not. At $n = 15$ a single observation moves the point estimate by more than half its value, so no confident statement about effect size is available, and the retained value cannot be discriminated from a unit error.

4.7. Confounding. The limitation that matters most is structural rather than distributional, and no amount of robustness checking touches it. The two groups compared differ in far more than their AI use, and the test conditions on nothing at all.

Pearl’s back-door criterion states the requirement precisely [16]. In a causal graph G , a set Z satisfies the back-door criterion relative to an ordered pair (X, Y) if

- (i) no node in Z is a descendant of X , and
- (ii) Z blocks every path between X and Y that begins with an arrow into X .

When such a Z exists and is observed, the interventional distribution is identified from observational data by the adjustment formula

$$(2) \quad \mathbb{P}(y \mid do(x)) = \sum_z \mathbb{P}(y \mid x, z) \mathbb{P}(z),$$

and a comparison of groups within levels of Z estimates a causal effect.

The two-sample t -test is exactly the case $Z = \emptyset$, and $\emptyset$ satisfies the criterion only if there is no back-door path at all, that is, no common cause of AI use and study time. That is implausible here, and this survey’s own data show why. Workload and course demands plausibly raise both AI use and total study hours, opening $X \leftarrow \text{workload} \rightarrow Y$; no workload item was collected. Prior ability and motivation plausibly do the same, and the **confidence** item, a weak proxy for them, correlates with **ai_hours** at $r = -0.432$, which is the signature of an open path. Year of study and course cluster plausibly affect both, and the sample is markedly unbalanced on both. Every one of these paths is left open, so the difference in Table 4 confounds the effect of AI use with the effects of whatever else distinguishes heavy from light users.

Adjustment is not available here for two reasons. The methods used are bivariate and cannot accommodate covariates, so no multivariable model can be fitted. More fundamentally, the two additional variables that were collected are unfit for the purpose. The **dependence** item, agreement with “I would struggle academically without AI tools,” is plausibly a *consequence* of AI use rather than a pre-treatment cause of it, which makes it a descendant of X and a mediator on the causal path. Conditioning on it would violate condition (i) directly, block part of the very effect being estimated, and can additionally induce collider bias if the mediator shares an unobserved cause with Y ; this is the “bad control” pattern catalogued by Cinelli *et al.* [3], and Section 5.4 demonstrates it numerically. Adjusting for the available covariate would therefore make the estimate worse, not better.

Measurement error compounds the problem in a way adjustment could not fix even if covariates were available: classical non-differential error in a dichotomised X attenuates the observed contrast toward zero, so part of the non-significance may be measurement artefact rather than absence of association.

The consequence for wording is strict. What has been established is that *in this sample, no statistically significant difference in mean independent study time was observed between low and high AI-use students*. It has not been established that AI use does not affect study time, nor that the observed difference estimates any causal quantity.

5. SECONDARY ANALYSIS: BACK-DOOR ADJUSTMENT

5.1. Motivation and dataset selection. Section 4.7 leaves two claims untested: that a valid adjustment set would change the answer, and that adjusting for a mediator would damage it. Neither can be checked on fifteen respondents with no confounders measured, so the same causal question is put to a public dataset that has the variables this instrument lacks.

Four candidates were assessed against the requirements of Section 3. The Middlebury survey [4] has the best variable match, including weekly study hours and GenAI frequency for 634 students, but respondent-level microdata is not distributed. The FLoRA dataset [11] supplies fine-grained interaction logs, but its temporal fields are event timestamps within a two-week assignment and measure time on task, not a weekly total. The global ChatGPT survey [15] has 22,963 respondents across 120 countries, but a full-text search of its questionnaire returns no quantitative study-time item, only Likert perception items; sample size cannot rescue a variable the instrument never asked for.

The Colombian study-habits survey [14] survives: 357 undergraduates at the Manizales campus of a Colombian public university across 14 programmes, with complete cases on every required variable, 357 of 357.

5.2. Graph and adjustment set. The assumed graph G has node set $\{X, Y, W, A, L, M_1, M_2\}$ and edge set

$$W \rightarrow X, \quad W \rightarrow Y, \quad A \rightarrow X, \quad A \rightarrow Y, \quad L \rightarrow X, \quad X \rightarrow M_1, \quad M_1 \rightarrow Y, \quad X \rightarrow M_2, \quad M_2 \rightarrow Y.$$

Here W is competing time demands, A collects prior ability, motivation and course difficulty, L is AI literacy and access, M_1 is perceived help understanding topics, and M_2 is AI-induced procrastination.

Since W , A , M_1 and M_2 each have degree two with neighbours exactly $\{X, Y\}$, and L has degree one, every simple X – Y path has length two and no path routes through L . The enumeration in Table 7 is therefore complete by construction rather than by sampling.

Path	Back-door?	Blocked by $Z = \{W, A\}$?	Blocked by $Z' = \{W, A, M_1\}$?
$X \leftarrow W \rightarrow Y$	yes	yes, fork at $W \in Z$	yes
$X \leftarrow A \rightarrow Y$	yes	yes, fork at $A \in Z$	yes
$X \rightarrow M_1 \rightarrow Y$	no, causal	no, left open	yes, and that is the fault
$X \rightarrow M_2 \rightarrow Y$	no, causal	no, left open	no

TABLE 7. Complete enumeration of the four X – Y paths.

$Z = \{W, A\}$ satisfies the criterion. Neither element is a descendant of X , since $\text{de}(X) = \{M_1, M_2, Y\}$; both back-door paths are blocked at their fork; no element is a collider on any X – Y path, so conditioning opens nothing; and both causal paths remain open, so the estimand is the *total* effect with the efficiency and procrastination channels summed.

$Z' = \{W, A, M_1\}$ fails, and fails on condition (i) rather than condition (ii). The edge $X \rightarrow M_1$ places $M_1 \in \text{de}(X)$. Note that Z' still blocks every back-door path, which is exactly why the failure must be diagnosed by (i): M_1 is a mediator, not a confounder.

L lies on no X – Y path, so Z blocks all back-door paths whether or not L is adjoined, and L is omitted. Adjoining it would not be invalid, but L is instrument-like and including instruments can amplify bias from unmeasured confounding [3]. The omission rests on an exclusion restriction: L influences Y only through X . If the true graph contains $L \rightarrow Y$, or a latent U with $L \leftarrow U \rightarrow Y$, then $X \leftarrow L \rightarrow Y$ is an open back-door path and $Z = \{W, A\}$ is insufficient.

The set A is operationalised as three sub-nodes rather than one composite: cumulative GPA, procrastination frequency, and academic programme entered as 13 dummies. This refinement leaves the criterion intact, because each sub-node retains degree two with neighbours $\{X, Y\}$, so the single fork through A is replaced by three parallel forks, each blocked at its middle node. The invariance is conditional and not free: omitting even one sub-node leaves that fork unblocked.

5.3. Estimation and results. Y is recorded as six ordered, interval-censored bands with an open top category, so the primary estimator is proportional-odds ordinal logistic regression, with ordinary least squares on band midpoints and HC3 standard errors reported alongside as the only reading interpretable in hours. The right-hand side is identical for both: X , W , and the three sub-nodes of A . This regression adjustment realises (2) under a linear-index restriction, and M_1 and M_2 are excluded so that both causal paths stay open.

Estimator	$\hat{\beta}_X$	SE	95% CI	statistic	p
Ordinal logit (log-odds per step)	0.2158	0.1319	(−0.0428, 0.4744)	$z = 1.635$	0.102
OLS on midpoints, HC3 (h/week)	0.1589	0.0944	(−0.0260, 0.3439)	$t = 1.684$	0.092
OLS, unadjusted (h/week)	0.0883	0.0965		$t = 0.915$	0.360

TABLE 8. Back-door-adjusted estimates under $Z = \{W, A\}$, $n = 357$, with the unadjusted estimate for reference. Odds ratio for the ordinal fit is $e^{0.2158} = 1.241$.

Neither adjusted estimate rejects $H_0 : \beta_X = 0$ at $\alpha = 0.05$. Both point estimates are positive: adjusting for competing time demands and ability, more frequent AI use is associated with *more* independent study time, about +0.16 hours per week per step on the five-point frequency scale, which is the opposite sign to the substitution hypothesis.

Adjustment for Z nearly doubles the point estimate relative to the unadjusted comparison, from 0.088 to 0.159 hours. The unadjusted figure was not merely imprecise; it was biased toward zero by the confounders. This is the concrete evidence that the confounding described in Section 4.7 is material and not

hypothetical, and it is what the t -test of Section 4 is structurally unable to see.

One caveat attaches to reading the ordinal coefficient as the quantity in (2). The adjustment formula returns a *marginal* interventional distribution, obtained by averaging $\mathbb{P}(y \mid x, z)$ over the marginal $\mathbb{P}(z)$, whereas the proportional-odds coefficient $\hat{\beta}_X$ is a *conditional* log-odds ratio holding Z fixed. The logistic link is not collapsible, so the two differ even when Z is a valid adjustment set and the model is correctly specified [6]. The linear model does not have this problem: the OLS coefficient on band midpoints is collapsible, so it is the row of Table 8 that maps directly onto (2). The ordinal fit remains primary because it does not require a ceiling to be invented for the open top band, but the hours-scale reading is the one entitled to the interventional interpretation.

5.4. Falsification: adjusting for the mediator. Refitting with $Z' = \{W, A, M_1\}$, on the same data with the same n and one variable added, gives Table 9.

Estimator	$\hat{\beta}_X$ under Z	$\hat{\beta}_X$ under Z'	Change	p under Z'
Ordinal logit	0.2158	0.1017	−52.9%	0.495
OLS, HC3 (h/week)	0.1589	0.0967	−39.2%	0.369

TABLE 9. Effect of conditioning on the descendant M_1 .

Mediator-inclusion bias is demonstrated rather than asserted. Conditioning on M_1 collapses the estimate toward zero in the direction the graph predicts, because path $X \rightarrow M_1 \rightarrow Y$ of Table 7 is blocked and only the M_2 channel remains in the coefficient. The p -value moves from 0.10 to 0.50. An analyst who “controlled for everything available” would report a substantively different and non-identified result. This is the empirical form of the argument in Section 4.7 for not adjusting on **dependence**.

5.5. Limitations of the secondary analysis. The outcome’s top band is open-ended and holds 51.26% of the sample, so the midpoint recode must assume a ceiling. Coding it at 5.5, 6.5 and 7.5 hours gives $\hat{\beta}_X = 0.1589$ ($p = 0.092$), 0.1906 ($p = 0.109$) and 0.2222 ($p = 0.127$). The sign and the non-significance are stable and the magnitude scales with the assumed ceiling, which is why the ordinal model, which makes no such assumption, is primary.

W is a partial proxy. The instrument’s only competing-demands item distinguishes full-time students from those who study and work, and captures no assignment-load information, so residual workload confounding is not adjusted away and the requirement that Z be measured without error fails.

The procrastination sub-node of A may be post-treatment. The item asks about procrastination in general and does not separate the trait, a pre-treatment cause, from AI-induced procrastination, which is M_2 and a descendant of X . To the extent it measures the latter, including it violates condition (i), the same error demonstrated in Section 5.4.

The graph is asserted from domain reasoning and is not verified from the data; the sample is a single Colombian campus; and the design is cross-sectional and self-reported. What Table 8 reports is an adjusted association that equals a causal effect only under the assumption of no unmeasured confounding.

6. CONCLUSION

The pooled two-sample t -test gives $t = -1.4944$ on 13 degrees of freedom against a critical value of ± 2.1604 , with $p = 0.1590$, so $H_0 : \mu_1 = \mu_2$ is not rejected at $\alpha = 0.05$. Theorem 1 shows the critical value is exact under the stated model, and the median split does not compromise it. The decision survives every relaxation attempted: Welch’s test, an exact permutation test free of parametric assumptions, the Mann–Whitney test, and deletion of the most influential observation all agree (Table 6).

The result should be recorded as inconclusive rather than null, for reasons quantified above rather than merely listed. The confidence interval $(-17.47, 3.18)$ hours per week excludes no effect of practical interest. The smallest difference the design could have detected with probability 0.80 is about 14.5 hours per week, larger than the average amount of studying being measured, so failure to detect was the likely outcome regardless of the truth. The point estimate moves by half its value when one respondent is removed. The sample is a self-selected convenience sample of fifteen with no calculable

inclusion probabilities, so the inferential machinery has no design basis. Both variables are self-reported recall estimates with an ambiguous response definition and at least one probable unit error, and three respondents demonstrably did not read “AI tool use” as the instrument intended. And the test conditions on the empty set, which does not satisfy the back-door criterion in any plausible graph for these variables.

That last point is the one the secondary analysis makes concrete. On 357 Colombian undergraduates, adjusting for a set that provably satisfies the criterion moves the estimate from +0.088 to +0.159 hours per week, while adjusting instead for a post-treatment mediator moves it to +0.097 and the p -value from 0.10 to 0.50. The difference between a valid and an invalid adjustment set is larger than the effect being estimated. A bivariate test that conditions on nothing cannot report a causal quantity, however many robustness checks it passes.

Both samples point the same way, away from substitution: high AI users in the primary sample studied 7.1 hours per week more on average, and the adjusted secondary estimate is +0.159 hours per week per step of AI-use frequency. Neither reaches significance, they differ in country, scale, measurement and adjustment, and the primary estimate is far too unstable to quantify. Two samples agreeing is mildly suggestive and nothing more.

6.1. Returning to the survey objective. Measured against the three aims set out in Section 1, the study achieved the first and the third and was defeated on the second by its own sample size. Weekly AI-tool use and independent study time were measured and fully described for all fifteen respondents (Tables 2 and 3). The hypothesis test was formulated, justified and carried out correctly, but returned a decision that the interval estimate shows to be uninformative. The limits of that conclusion were established in detail, and they are the substance of what this project found.

The exercise touched most of the inferential apparatus of the module and it is worth naming where. *Sampling methods* appear in Section 3.3, where the absence of a probability design removes the warrant for treating $\bar{y}$ as unbiased for μ — the point at which a convenience sample stops being merely inconvenient and starts undermining the inference itself. *Hypothesis testing* appears in Section 4: a two-tailed $H_0 : \mu_1 = \mu_2$ at $\alpha = 0.05$, a pooled two-sample t statistic chosen because σ is unknown and the samples are small, a critical region of ± 2.1604 on 13 degrees of freedom, and a decision not to reject. *Assumptions* are checked rather than assumed in Section 4.6, where normality is rejected in one group and homoscedasticity is left unverified, which is why four robustness analyses are reported alongside the primary one. *Confidence intervals* appear in Corollary 3 and carry more information than the test does: the interval $(-17.47, 3.18)$ is what turns “not significant” into “not informative”. *Type I and type II errors* are separated in Section 4.5: the first is controlled exactly at 0.05 by Theorem 1, the second is not controlled at all, and expressing it as a minimum detectable effect of 14.5 hours per week makes the shortfall concrete in the units of the problem. *Limitations* run throughout rather than being confined to a closing paragraph, because at $n = 15$ they are not a caveat on the finding, they are the finding.

The single most useful lesson is one the module states and this dataset made unavoidable: failing to reject a null hypothesis is not evidence that it is true. The same $p = 0.159$ would have been produced by a real effect of seven hours per week and by no effect at all, and only the interval estimate distinguishes those situations.

6.2. Recommendations for a future iteration. Five changes would make a future iteration informative. Raise the sample to at least 28 per group, and preferably to the 55 per group implied by a pre-specified five-hour difference rather than by the observed effect. Replace convenience recruitment with probability sampling stratified by school and year, which would also correct the SOE and Year-3 over-representation. Measure time objectively where possible, through device screen-time reports or learning-management-system activity logs, rather than by recall. Collect pre-treatment confounders explicitly, namely assignment count, contact hours and prior grade point average, so that a set satisfying the back-door criterion can be formed and a multivariable model fitted, and analyse X on its original continuous scale rather than dichotomised, which recovers the power lost in Remark 2. Finally, pilot the questionnaire to repair the option lists that respondents overrode with free text, define “independent study” so that intermittent AI use has a clear home, and add range validation to reject implausible hour values at entry.

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